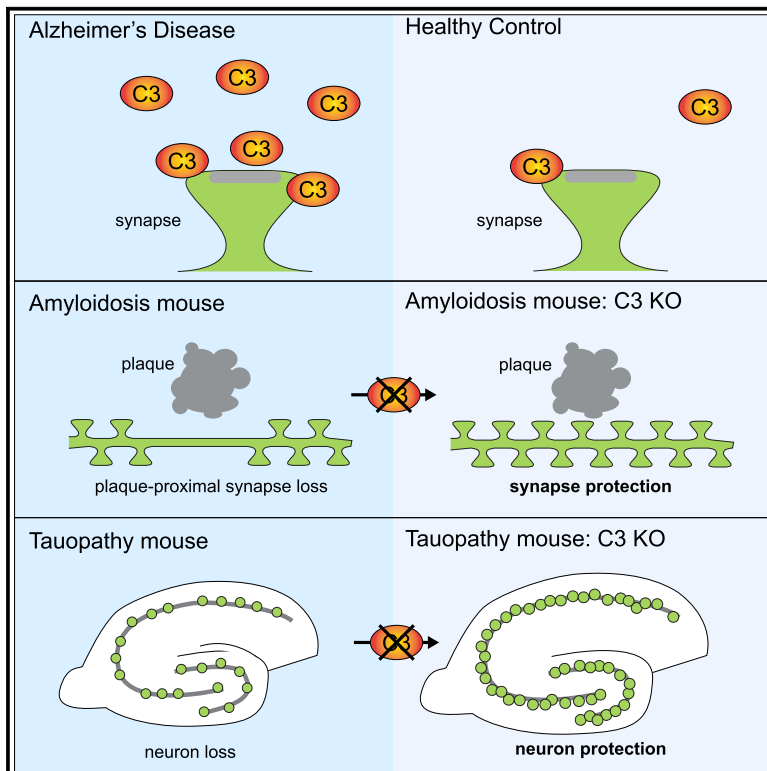


Cell Reports

Complement C3 Is Activated in Human AD Brain and Is Required for Neurodegeneration in Mouse Models of Amyloidosis and Tauopathy

Graphical Abstract



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In Brief

Wu et al. show that loss of the central complement component C3, which is elevated and activated in brains and cerebrospinal fluid from AD patients, ameliorates synapse loss and neurodegeneration in amyloidosis and tauopathy AD mouse models despite ongoing glial activation.

Highlights

- Increased glial classical complement expression in amyloidosis and tauopathy models
- C3 deficiency rescues plaque-proximal synapse loss in PS2APP mice
- C3 deficiency mitigates neurodegeneration and neuronal loss in TauP301S mice
- C3 protein is increased in brains and cerebrospinal fluid from AD patients



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Complement C3 Is Activated in Human AD Brain and Is Required for Neurodegeneration in Mouse Models of Amyloidosis and Tauopathy

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SUMMARY

Complement pathway overactivation can lead to neuronal damage in various neurological diseases. Although Alzheimer's disease (AD) is characterized by β -amyloid plaques and tau tangles, previous work examining complement has largely focused on amyloidosis models. We find that glial cells show increased expression of classical complement components and the central component C3 in mouse models of amyloidosis (PS2APP) and more extensively tauopathy (TauP301S). Blocking complement function by deleting C3 rescues plaque-associated synapse loss in PS2APP mice and ameliorates neuron loss and brain atrophy in TauP301S mice, improving neurophysiological and behavioral measurements. In addition, C3 protein is elevated in AD patient brains, including at synapses, and levels and processing of C3 are increased in AD patient CSF and correlate with tau. These results demonstrate that complement activation contributes to neurodegeneration caused by tau pathology and suggest that blocking C3 function might be protective in AD and other tauopathies.

INTRODUCTION

Alzheimer's disease (AD) is characterized by accumulation of extracellular amyloid plaques and intracellular neuronal tau tangles, as well as synapse and neuron loss (Jack et al., 2010; Spiess-Jones and Hyman, 2014). Amyloidosis can precede cognitive decline by decades in AD patients and does not correlate closely with neurodegeneration, whereas tau pathology correlates well with cognitive decline and synapse loss (Jack et al., 2010; Spiess-Jones and Hyman, 2014). Accumulating evidence

suggests that amyloidosis is upstream of tau and induces tau pathology, which then drives neurodegenerative processes (Choi et al., 2014; Jin et al., 2011; Mairet-Coello et al., 2013; Roberson et al., 2007; Zempel et al., 2010). Besides AD, tau is implicated in neurodegenerative diseases including frontotemporal dementia and progressive supranuclear palsy (Wang and Mandelkow, 2016), in which tau pathology can be present in the absence of amyloid pathology. Another prominent feature of AD is the presence of activated microglia and astrocytes, which may play protective roles early in disease but could also be synaptotoxic and damaging to neurons at later stages (Hansen et al., 2018; Lidde-low et al., 2017).

Components of the complement system, part of the innate immune system involved in clearance of pathogens and damaged cells, are expressed and secreted by microglia and astrocytes and can participate in synapse removal. This is particularly well studied in visual system development (Stephan et al., 2012), in which synapse pruning is mediated partly by microglial engulfment of synapses that are opsonized by complement, with knockout (KO) mice of C1q, C4, C3, and CR3 supporting a role for C3 activation in response to classical pathway initiation (Schafer et al., 2012; Sekar et al., 2016; Stevens et al., 2007). Research suggests that this developmental pathway is reactivated in models of amyloidosis and can contribute to synapse loss caused by β -amyloid (A β) (Fonseca et al., 2004; Hansen et al., 2018; Hong et al., 2016; Shi et al., 2015, 2017). In contrast to the subtle neurodegeneration phenotypes of amyloidosis mouse models, tauopathy models exhibit widespread neurodegeneration (Ittner et al., 2010; Roberson et al., 2007), resulting in brain atrophy that can be readily detected by MRI. However, whether complement activation mediates neuron loss in tauopathy models or relates to tau pathology in human AD is unclear.

Using RNA sequencing (RNA-seq) analysis of sorted brain cells and histological characterization of brain tissues, we show that mouse models of amyloidosis and of tauopathy independently induce expression of complement. Complement up-regulation was subtler and restricted to the vicinity of plaques



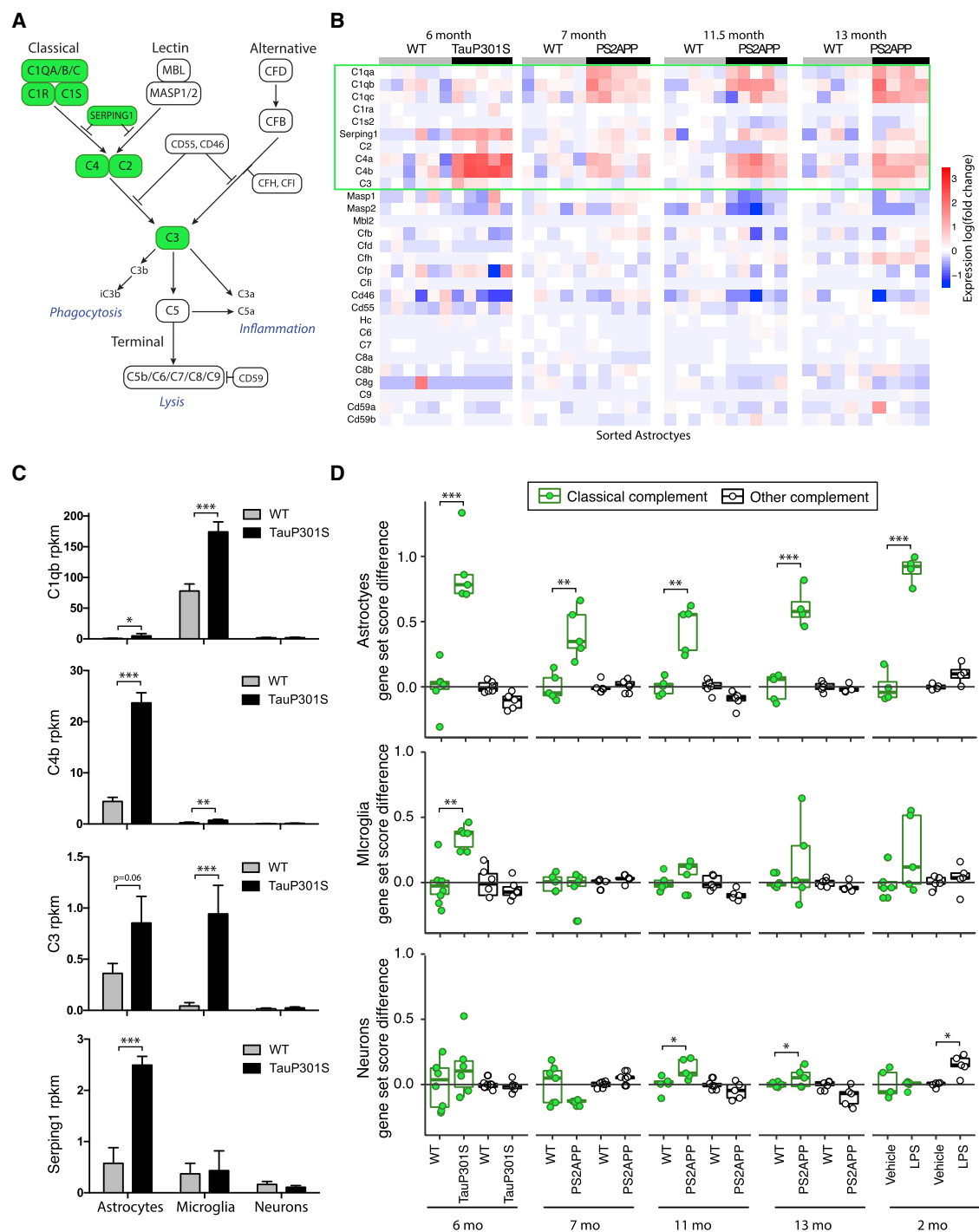


Figure 1. Expression of Classical Complement Genes in Sorted Astrocytes, Microglia, and Neurons from PS2APP and TauP301S Mice

(A) Simplified schematic of the complement pathway illustrating selected proteins. Proteins involved in the classical complement pathway are highlighted in green.

(B) Heatmap depicts relative expression levels of complement genes in sorted astrocytes from WT mice, 6-month-old TauP301S mice, and 7- to 13-month-old PS2APP mice. The expression $\log_2(\text{fold change})$ is with respect to the mean RPKM expression per gene in the WT samples for each cohort. The green frame indicates genes from the classical complement pathway.

(C) Expression of selected classical pathway genes in astrocytes, microglia, and neurons from WT and TauP301S mice. Data are shown as mean $\pm$ SEM.

(legend continued on next page)

in PS2APP mice but was more extensive in TauP301S mice, especially in the phospho-tau-filled hippocampus. Eliminating complement function using C3KO ameliorated synapse loss near plaques in PS2APP mice and reduced the frank neurodegeneration in TauP301S mice, which corresponded with functional benefits. We find that C3 protein accumulates in synapses in human AD brains with tau pathology. Furthermore, intact and processed C3 levels are increased in cerebrospinal fluid (CSF) from AD patients and correlate with tau levels, informing potential therapeutic strategies for targeting complement in AD.

RESULTS

Induction of Brain Complement Expression in PS2APP and TauP301S Mice

Acute insults can induce different states of astrocyte activation termed A1 (defined by a specific response to endotoxemia) and A2 (defined by a specific response to ischemia), with A1 astrocytes in particular upregulating several complement components (Liddel et al., 2017; Zamanian et al., 2012). We therefore wondered whether astrocytes in more chronic models such as tauopathy and amyloidosis might exhibit activation profiles that include increased expression of complement components. To investigate this, we sorted cells from the hippocampus of 6-month-old male TauP301S mice (Friedman et al., 2018), representing a disease stage with obvious tau pathology but before frank neurodegeneration (Yoshiyama et al., 2007), and analyzed datasets from sorted cells from PS2APP forebrain at various ages (7–13 months of age) that we had generated previously (Srinivasan et al., 2016). For comparison, we used datasets from lipopolysaccharide (LPS)-treated wild-type (WT) mice (Srinivasan et al., 2016), which have strongly induced reactive astrocytes and particularly strong induction of pan-reactive (induced by endotoxemia and ischemia) and A1 genes (Figure S1B). Although smaller in magnitude compared with LPS treatment, TauP301S astrocytes exhibited robust induction of A1 and pan-reactive genes (Figure S1). In comparison, PS2APP astrocytes exhibited relatively subtle astrocyte activation signatures at all tested ages (Figure S1B).

With regard to the complement pathway (Figure 1A), sorted astrocytes from TauP301S and PS2APP mice had significantly increased expression of genes involved in classical pathway activation (C1q through C3), but not genes more specific to the lectin, alternative, or terminal pathways of complement (Figures 1A and 1B). Although induction of classical pathway genes was most prominent in astrocytes, induction was also seen in microglia from TauP301S mice, but not microglia from PS2APP mice (Figures 1C and 1D). In contrast, expression of complement gene mRNAs in neurons was low and was not robustly changed in these disease models (Figures 1C and 1D). In general, classical complement induction at the mRNA level in astrocytes from TauP301S or PS2APP mice was comparable to or ap-

proached the level seen in LPS-treated mice, while genes from other parts of the complement system were not induced in these models (Figure 1D).

We next analyzed protein levels of the classical pathway initiating component, C1q, and the central complement component, C3, by immunohistochemistry (IHC) and compared the amyloid mice to the tauopathy mice. As previously reported (Dejanovic et al., 2018; Stephan et al., 2013), C1q staining is most abundant in the hippocampus in WT mice (Figure 2A). In PS2APP mice there was no obvious change in whole-brain or hippocampal C1q staining (Figures 2A and 2C), and C1q immunostaining was absent from areas occupied by plaques, as has previously been reported in some amyloid models (Fonseca et al., 2011) (Figure 2E). In contrast, TauP301S mice had significant elevation in whole-brain C1q staining and an even greater (~3-fold) increase in hippocampal C1q (Figures 2A and 2C). For C3, the most concentrated immunoreactivity co-localized with GFAP (glial fibrillary acidic protein; Figures 2B, 2F, and 2G). Although C3 transcript levels were comparable in microglia and astrocytes (Figure 1C), C3 immunoreactivity did not co-localize with microglia in WT or AD models (Figure S2), suggesting that astrocytes are the main producers of C3 protein in the brain (Lian et al., 2016). In PS2APP mice, there was a small but significant increase in astrocytic C3 staining, which was more pronounced when analysis was restricted to areas surrounding plaques (Figures 2D and 2F). In TauP301S mice, astrocytic C3 staining was robustly elevated in whole brain and particularly in hippocampus, a region that has strong tau pathology (see Figure S4), reflecting the increased abundance of reactive astrocytes (Figures 2D and 2G). Thus, TauP301S mice exhibit higher levels of classical complement proteins compared with PS2APP mice, with significantly increased neuropil deposition of C1q and a more robust and widespread increase in astrocytic C3 production.

C3KO Reduces Plaque-Associated Spine Loss in PS2APP Mice

Does complement deficiency affect synapse loss and neurodegeneration in models of tauopathy and amyloidosis? In aged PS2APP mice crossed to Thy1-GFP-M mice (which express GFP sparsely in pyramidal neurons), we previously observed dendritic spine loss that is restricted to the vicinity of plaques (Hanson et al., 2014). Here, we assessed the effects of C3KO on spine density along plaque-proximal and plaque-distant dendrites. In the WT background, C3KO had no effect on spine density or spine size (Figures 3A–3C). In PS2APP mice, there was ~50% reduced spine density near plaques, and this reduction in spine density was significantly rescued by C3KO (Figures 3A–3C), supporting the concept that complement mediates synapse loss in response to amyloidosis. In agreement with previous studies, we observed a significant increase in plaque number and size in C3KO mice (Shi et al., 2017) (Figures 3D–3F). Although C3 deficiency largely protected synapse number near

(D) Differences in gene set score expression of classical complement genes (green data points) and all other complement pathway genes (black data points) in animal models are indicated. The y axis reflects a difference in the gene set score in a given sample compared with the mean of the corresponding WT or vehicle samples. Boxes represent the interquartile range (IQR), and whiskers extend to the largest and smallest data points within 1.5 of the IQR from the box edges. Each point represents data from one mouse. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

(C) Unadjusted values from voom+lrimma differential expression analysis. (D) Two-way ANOVA with Tukey's post hoc test. See also Figure S1.

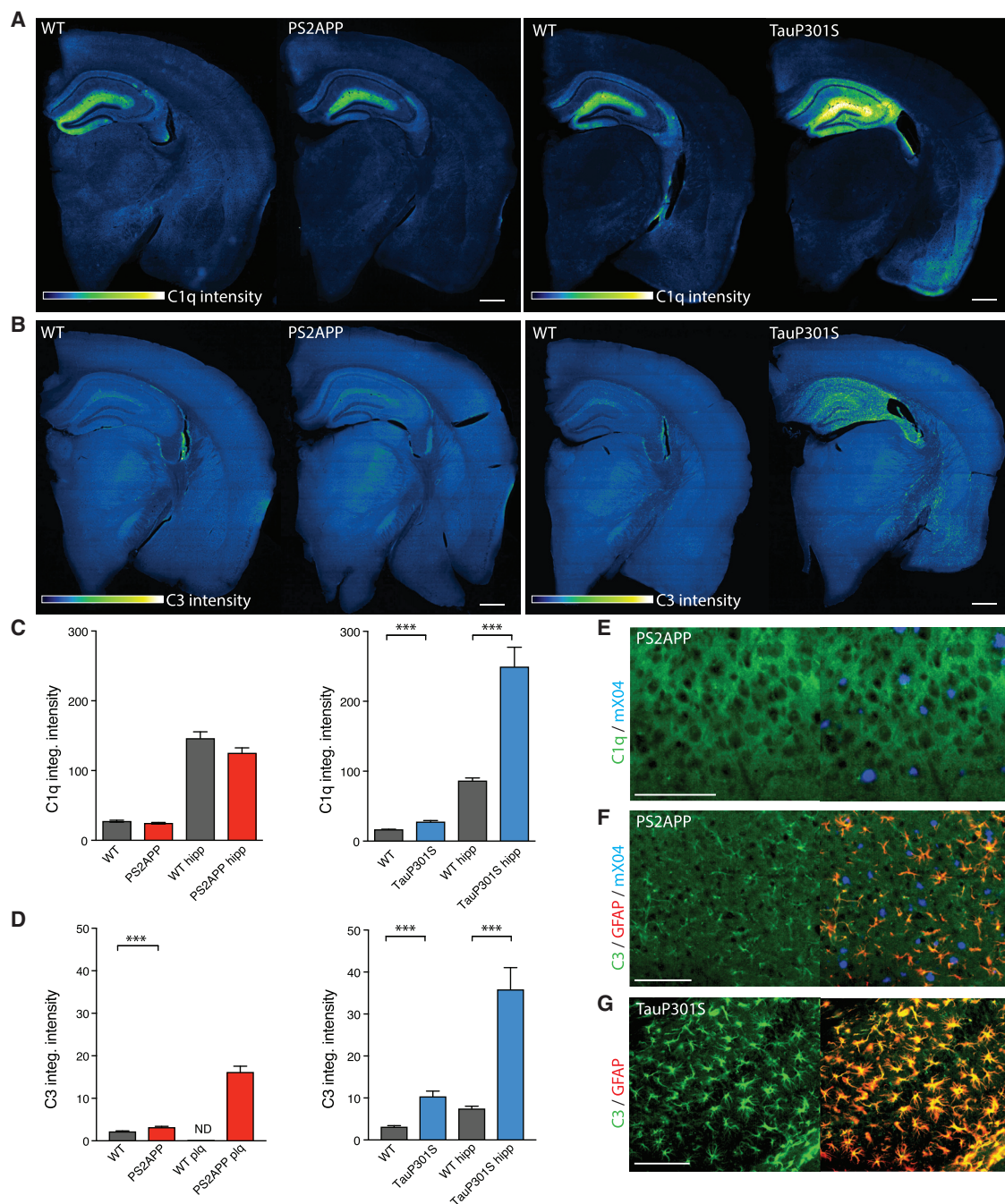


Figure 2. C1q and C3 Immunostaining in PS2APP and TauP301S Mice

(A and B) Representative images of C1q (A) and C3 (B) immunostaining in PS2APP and TauP301S mice and WT littermates at ages featuring widespread plaque (6-month-old PS2APP) and pTau pathology (9-month-old TauP301S). Scale bar, 500 μ m.

(C) Quantification of C1q immunofluorescence intensity in whole-brain sections and hippocampi of PS2APP and TauP301S mice.

(D) Quantification of astrocytic C3 immunofluorescence intensity. Analysis of C3 staining was restricted to regions defined by a mask of the GFAP staining and is shown for whole sections versus regions within a dilation of 13.8 μ m from plaques for PS2APP mice and for whole sections versus hippocampi for TauP301S mice.

(E) Example immunostaining for C1q and methoxy-X04 in the CA1 region of the PS2APP model.

(F and G) Example immunostaining for C3, GFAP, and methoxy-X04 in PS2APP (F) and TauP301S (G) mice. Scale bar, 100 μ m.

n = 15–21 mice per genotype with 2 sections per mouse. ***p < 0.001. Unpaired Student's t test. All data are presented as mean $\pm$ SEM. See also Figure S2.

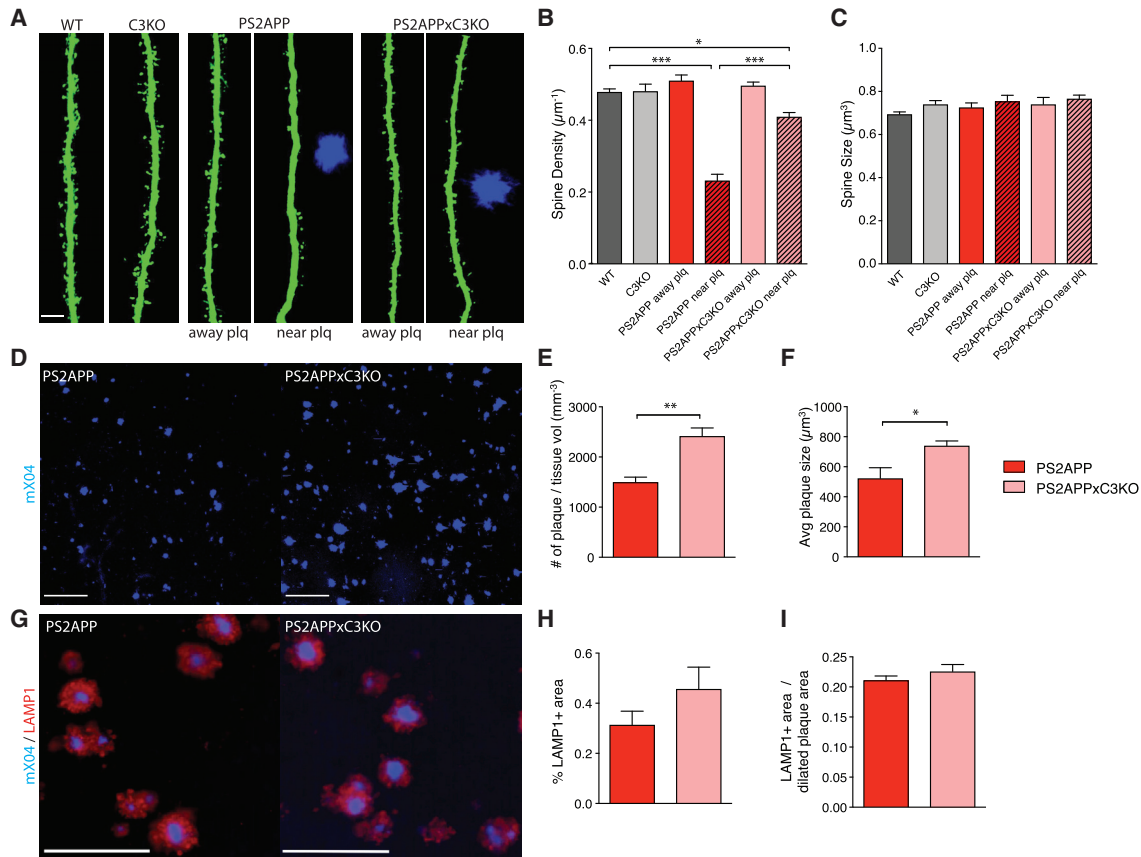


Figure 3. C3 Deletion Rescues Plaque-Associated Spine Loss in PS2APP Mice and Increases Plaque Load

(A) Representative images of GFP-expressing neurons in 6-month-old male mice with genotypes as indicated. Plaque was stained blue with methoxy-X04. “Near plq” is defined as plaque within 20 μm of dendrite, and “away plq” is defined as no plaque within 100 μm of field of view. Scale bar, 30 μm . (B and C) Analysis of spine density (B) and volume (C). $n = 5$ mice per genotype with 3–5 dendrites per mouse. (D) Representative images of methoxy-X04-stained plaques in the somatosensory cortex of PS2APP and PS2APPxC3KO mice. Plaques were imaged at 20 $\times$ and a 300 μm depth. (E and F) Analysis of plaque density (E) and plaque size (F) from the same mice used for spine analysis. (G) Representative images of methoxy-X04-immunostained plaques and LAMP1 in brain sections of PS2APP and PS2APPxC3KO mice. (H) Percentage of LAMP1-positive area in brains of PS2APP and PS2APPxC3KO mice. (I) Fraction of LAMP1-positive area within a dilation of 13.8 μm from plaques in PS2APP and PS2APPxC3KO mice. $n = 15$ –21 mice per genotype with 2 sections per mouse. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (B and C) One-way ANOVA with Tukey’s post hoc test. (E, F, H, and I) Unpaired Student’s t test. All data are presented as mean $\pm$ SEM. Scale bars, 100 μm . See also Figure S3.

plaque, the total area of dystrophic neurites (as measured by LAMP1 immunohistochemistry) (Gowrishankar et al., 2015) increased by $\sim 50\%$ and correlated with the greater plaque burden in PS2APPxC3KO mice, while the LAMP1+ area per individual plaques was unchanged (Figures 3G–3I), indicating that lack of C3 does not affect the phenotype of plaque-associated dystrophic neurites. To further examine plaque-related pathology in these mice, we used Campbell Switzer (CS) silver stain and stained for astrocytes and microglia using GFAP and Iba1, respectively. Consistent with an increased plaque load, we observed a trend toward higher CS staining in whole-brain sections (Figures S3A and S3B). Although the overall GFAP and Iba1 area was significantly increased in PS2APP versus WT brains (Figures S3C, S3E, S3F, and S3H), there were no differences in overall or plaque-associated GFAP or Iba1 staining in

PS2APPxC3KO mice (Figures S3C–S3H), indicating that C3 did not significantly affect gliosis in this model.

C3KO Reduces Brain Atrophy and Neuron Loss in TauP301S Mice

We next tested whether C3KO could reduce neurodegeneration in tauopathy mice. Using longitudinal volumetric MRI (vMRI) measurements at 3, 6, and 9 months of age, we examined the effect of C3KO on brain volume in TauP301S mice. Although all volumetric measurements were similar in 3- and 6-month-old animals across genotypes, brain atrophy developed in TauP301S mice between 6 and 9 months of age, as indicated by rising ventricular volumes and declining cortical and hippocampal volumes (Figures 4A and 4B). C3KO did not significantly alter the trajectory of brain volume changes in

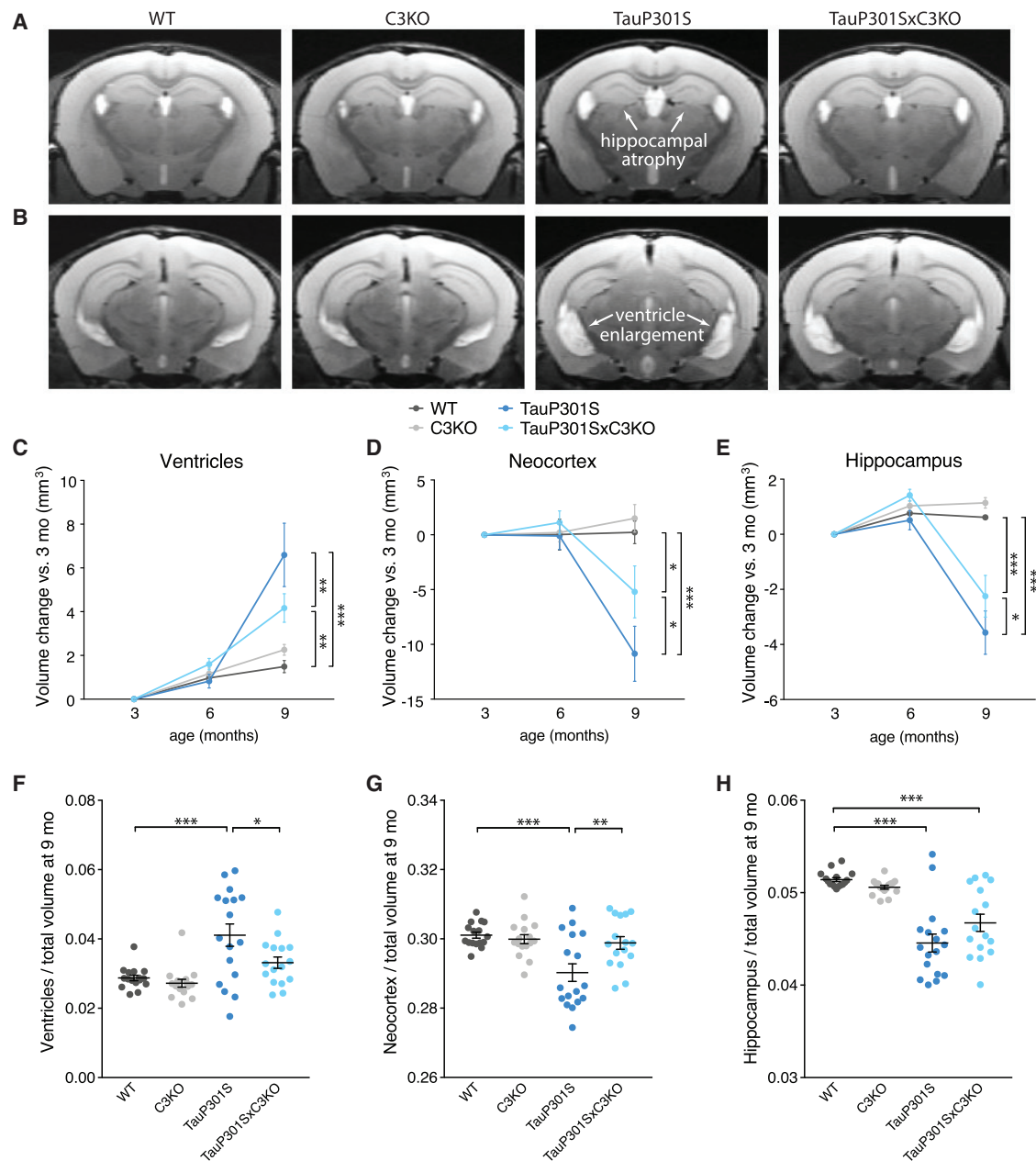


Figure 4. C3 Deletion Ameliorates Brain Atrophy in TauP301S Mice

(A and B) Representative volumetric MRI images in male mice at 9 months of age illustrating changes in hippocampal and ventricular volumes. Slices were selected at levels at which hippocampal atrophy (A) and ventricle enlargement (B) are evident in TauP301S mice (indicated by arrows).

(C–E) Longitudinal volumetric MRI quantification shows changes in the volume of ventricles (C), neocortex (D), and hippocampus (E) at 6 and 9 months compared with 3 months of age in genotypes as indicated. $n = 14$ mice per genotype (equal numbers of males and females combined).

(F–H) Ventricular (F), neocortical (G), and hippocampal (H) volumes normalized to total brain volume in two cohorts of 9-month-old male mice. Volumetric measures from male mice from the first cohort (C–E) were combined with data from a second cohort.

$n = 16$ – 17 mice per genotype. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (C–E) Two-way ANOVA with Tukey's post hoc test. (F–H) One-way ANOVA with Tukey's post hoc test. All data are presented as mean $\pm$ SEM. See also Figure S4.

WT mice (Figures 4C–4E). However, C3KO significantly ameliorated the increase in ventricular volume and the decline in cortical and hippocampal volume in TauP301S mice at 9 months (Figures 4A–4E). To assess reproducibility of these results, we obtained vMRI data from a second cohort of mice

at 9 months of age. Analysis of the combined cohorts indicated that both atrophy in TauP301S mice and partial rescue by C3KO at 9 months of age were mainly driven by male mice (Figures 4F–4H), with female mice exhibiting only subtle effects of C3KO (Figure S4). Therefore, in subsequent analysis of tissue

from these mice, we analyzed male and female mice separately.

We next examined neurodegeneration by histology in 9-month-old male mice, focusing on the hippocampus. In TauP301S hippocampi, Amino-Cupric (AminoCu) staining that detects neuronal degeneration was increased, NeuN staining for surviving neurons was decreased, and hippocampal area was reduced, suggesting that neurons are lost at 9 months of age in these mice (Figures 5A–5E). Corroborating the vMRI data, hippocampi of TauP301SxC3KO mice had significantly ameliorated loss of NeuN staining and trended toward reduced AminoCu staining and hippocampal area loss (Figures 5A–5E). Altogether, these data show that C3 deficiency is protective against neurodegeneration in TauP301S mice. This partial rescue of neurodegeneration was not associated with significant reduction in the extent of tau pathology or prevention of gliosis, because AT8 staining was unchanged and only trends toward reduced GFAP and Iba1 staining were detected (Figures S5A–S5F). Mirroring the level of gliosis, the concentration of many cytokines was highly increased in TauP301S compared with WT and C3KO brain lysates, and no significant reductions in cytokine levels in TauP301SxC3KO mice compared with TauP301S mice could be detected (Figure S5G). In contrast to male mice, female TauP301S mice in this cohort showed subtler neurodegeneration at 9 months of age, and effects of C3KO could not be resolved (Figures S5H–S5M).

Because activated microglia can engulf complement-tagged synapses (Dejanovic et al., 2018; Hong et al., 2016; Schafer et al., 2012), we assayed microglial engulfment of synapses in 9-month-old male TauP301S mice by measuring the number of synapsin puncta detected inside CD68-positive microglial lysosomes by 3D reconstruction of confocal z stacks (Figure 5F). The quantity of synapsin puncta within microglial lysosomes was significantly increased (~3-fold) in TauP301S mice compared with WT mice. In TauP301SxC3KO hippocampi, there was less microglial synapse engulfment compared with TauP301S mice, although the difference was not significant (Figures 5F and 5G). The decrease in synapsin puncta inside CD68 structures was mainly driven by decreased CD68 volume per microglia in TauP301SxC3KO versus TauP301S hippocampi (Figures 5F–5H). Altogether, the rescue of brain atrophy by MRI, rescue of neuron loss by IHC, and trend toward reduced markers of degeneration and synapse engulfment indicate the neuroprotective effects of C3KO in this tauopathy model.

C3KO Partially Normalizes Neurophysiological and Behavioral Alterations in TauP301S Mice

The reduction of brain atrophy and neuron loss in TauP301SxC3KO mice could reflect (1) less beneficial removal of dead or dying neurons and debris or (2) less harmful removal of viable synapses and neurons by complement activation. To test for effects of C3KO on synaptic and neuronal function in TauP301S mice, we measured population spike (PS) long-term potentiation (LTP), which is robustly impaired in TauP301S mice at stages before hippocampal atrophy. C3KO alone did not affect PS LTP in brain slices from 6-month-old WT male mice, whereas C3KO ameliorated PS LTP deficits in TauP301S

mice (Figures 6A and 6B), demonstrating partial normalization of neurophysiological abnormalities.

To test for behavioral effects of C3KO, we next measured locomotor activity by measuring the total beam breaks of 9-month-old male mice (Figure 6C) and female mice (Figure S6A) in an open field. Male TauP301S mice showed hyperactivity in this assay, like many AD mouse models, and the hyperactivity was significantly decreased in TauP301SxC3KO mice (Figures 6C and 6D). Mice lacking C3 alone showed no difference in activity from that of WT mice (Figures 6C and 6D). In line with only subtle neurodegeneration in this cohort, female TauP301S mice did not exhibit robust hyperactivity (Figures S6A and S6B). To try to more directly measure spatial learning and memory, male mice were analyzed in the Morris water maze test (Figures S6C and S6D). However, TauP301S mice from this cohort did not show robust impairment in this behavioral readout, and the effect of C3KO on cognitive impairment could not be assessed (Figures S6C and S6D). Altogether, the electrophysiological and behavioral analyses suggest that complement inhibition in TauP301S mice is neuroprotective and leads to partial normalization of functional readouts.

Synaptic C3 Levels Are Increased in AD Brains, and Increased Intact and Processed C3 Can Be Detected in CSF

Given that C3 tagging of synapses has been shown to lead to synapse loss in amyloidosis models (Shi et al., 2017), we next analyzed whether C3 is present at synapses in human AD brains. For these experiments, we performed biochemical purification of postsynaptic densities (PSDs) from superior frontal gyrus (SFG) samples from control, early AD (EAD), and AD brains (Figures 7A and 7B). Confirming the more advanced stage of disease, phospho-tau (pTau; AT8 epitope) could be readily detected in lysates from AD, but not EAD, brain samples (Figure 7C). Although C3 levels in EAD lysates and PSDs were not significantly different from controls, there was a trend toward increased C3 levels in AD lysates and significantly augmented C3 levels in PSDs from AD patients (Figures 7A and 7B). Thus, C3 levels are increased, especially at synapses, and in particular at a stage of AD featuring tau pathology.

We next wondered whether activated, processed (i.e., cleaved) C3 is detectable and changed in biofluids from AD patients. Using a highly sensitive assay on the Simoa platform (Figures S7A–S7C; see also STAR Methods), we measured the levels of intact C3 or processed C3 (C3b, iC3b, and C3c) in CSF from cognitively normal and AD patients (Figure 7; Table S1). Intact and processed C3 were ~1.5-fold and ~2.5-fold increased, respectively, in CSF from AD patients (Figures 7D and 7F). The ratio of processed/intact C3 was also significantly elevated (Figure 7H), suggesting that complement expression and activation are elevated in AD brains. When normal and AD samples were analyzed together, intact and processed C3 levels, as well as the ratio, correlated significantly with CSF levels of tau (Figures 7E, 7G, and 7I). To confirm this finding, we analyzed an independent, second cohort of CSF samples from normal and AD patients (Figures S7D–S7I; Table S1). Similar to the first cohort, intact and processed C3 levels were both significantly increased in CSF from AD patients and significantly

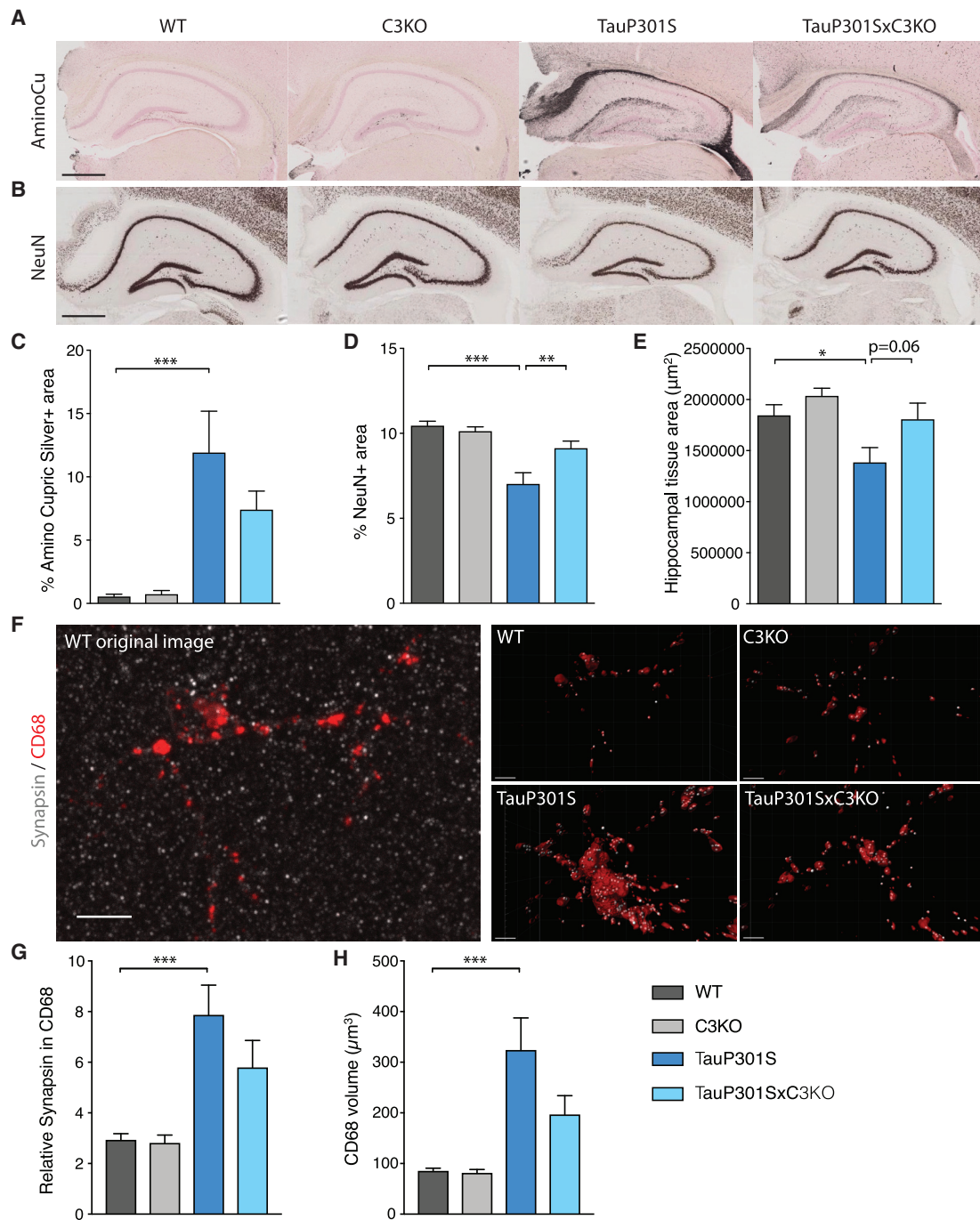


Figure 5. Effects of C3KO on Neuronal Degeneration Markers and Microglial Synapse Engulfment in TauP301S Mice

(A and B) Representative images of hippocampi stained with AminoCu (A) and NeuN (B) from 9-month-old male mice, with genotypes as indicated. Scale bar, 500 μm .

(C and D) Percentages of AminoCu- (C) and NeuN- (D) covered hippocampal area are shown.

(E) Total hippocampal area is shown. $n = 9$ –10 mice per genotype with 8 sections per mouse.

(F) Maximum intensity projection of confocal z stack and 3D reconstructions of synapsin (white) in CD68-stained microglial lysosomes (red) in hippocampal CA1 region from 9-month-old male mice. Scale bar, 5 μm .

(G) Relative number of synapsin puncta in CD68-stained structures per microglia.

(H) Volume of CD68+ structures in genotypes are indicated. $n = 8$ –10 mice per genotype with 7–8 microglia per mouse.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (C–E) One-way ANOVA with Dunnett's post hoc test. (G) One-way ANOVA with Tukey's post hoc test. All data are presented as mean $\pm$ SEM. See also Figure S5.

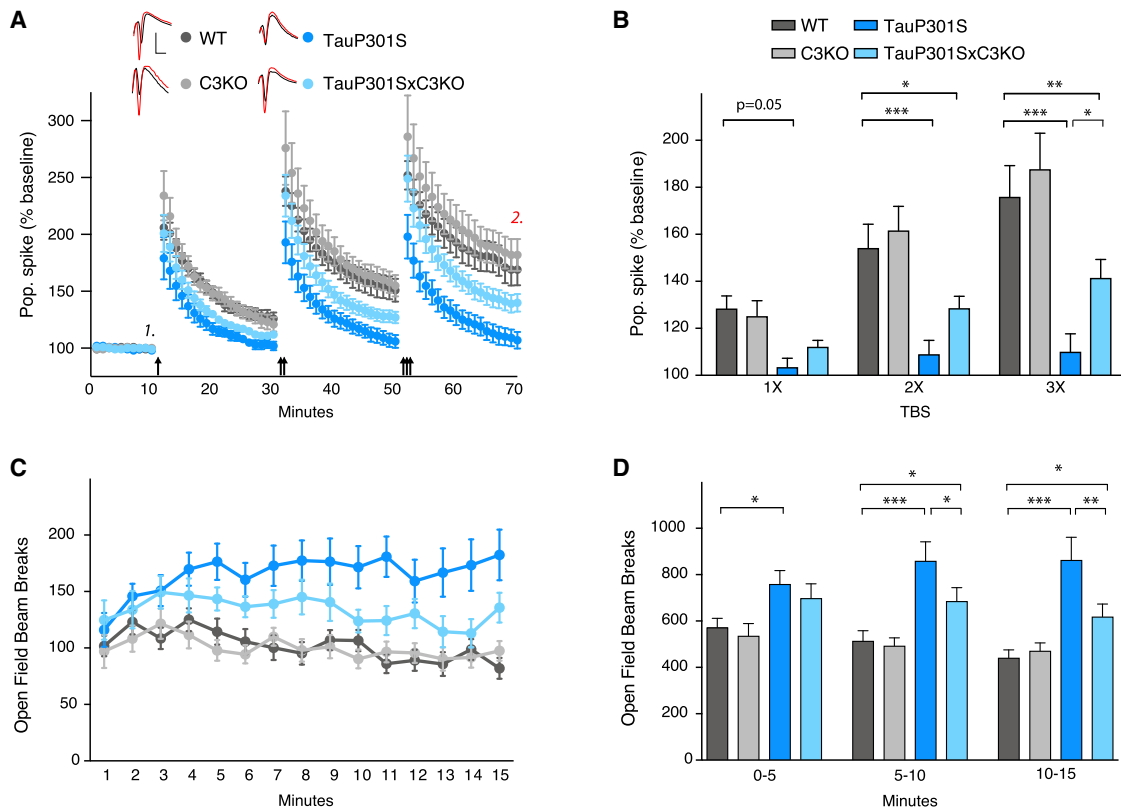


Figure 6. C3 Deletion Partially Rescues Neurophysiological and Behavioral Abnormalities in TauP301S Mice

(A and B) Population spike long-term potentiation (LTP) in hippocampal area CA1 was measured using field recordings from the cell body layer in brain slices from 6-month-old male mice. LTP was induced using 1, 2, or 3 bouts of theta burst stimulation (TBS) at different times in the recording (as indicated by arrows, A) and quantified as the average percentage increase in population spike magnitude during the last 5 min of each portion of the experiment (B). $n = 5$ mice per genotype, insets show example traces overlaid from time 1 (black) and time 2 (red) as indicated on the chart, and scalebars are 1 mV and 5 mS.

(C) Spontaneous locomotor activity of 9-month-old male mice in a 15-min open-field session as measured by total beam breaks per minute.

(D) Total beam breaks in 5-min intervals.

$n = 18$ – 20 mice per genotype. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. (B and D) Two-way ANOVA with Fisher's least significant difference (LSD) test. All data are presented as mean $\pm$ SEM. See also Figure S6.

correlated with tau levels, while changes in the ratio were not significant in the second cohort (Figures S7D–S7I). The correlation of tau levels with intact and processed C3 levels was consistent in both cohorts, but this could be contributed to by differences in tau levels between AD patients and controls, and future studies examining large numbers of AD patients will be valuable in more closely examining relationships between intact and processed C3 levels and various biomarkers, including tau. Altogether, the human brain lysate and CSF data strongly suggest that complement C3 is activated in the CNS of AD patients.

DISCUSSION

Although amyloidosis and tauopathy are implicated in AD neurodegeneration, the exact underlying molecular mechanisms and pathways remain poorly understood. Mounting evidence in cells and AD mouse models suggest that A β acts upstream of tau and induces tau-mediated neurotoxicity (Choi et al., 2014; Ittner et al., 2010; Lewis et al., 2001; Rapoport et al., 2002; Roberson et al., 2007). Here, we show that independently from amyloid-

osis, tauopathy can induce classical complement gene expression in glial cells and that complement protein expression is strongest in the hippocampus, where tau pathology is most prominent. C1q, which can localize to synapses and accumulates with aging (Stephan et al., 2013), showed elevated neuropil staining in TauP301S mice, but not PS2APP mice, despite increased C1q gene expression in both models. This suggests accumulation, rather than production, might drive increased neuropil C1q levels and that TauP301S mice may more broadly express C1q receptors that promote deposition. Nonetheless, that C3KO rescued plaque-associated synapse loss indicates synaptotoxic A β can trigger complement activation in PS2APP mice even in the absence of widespread C1q accumulation. In addition to broader C1q deposition, the more extensive astrocyte activation and elevated C3 protein expression in TauP301S mice might contribute to the stronger neurodegeneration phenotype in TauP301S versus PS2APP mice. These results are consistent with the more widespread neurodegeneration in tauopathy models versus the minimal neurodegeneration in amyloidosis models in general (Götz et al., 2018).

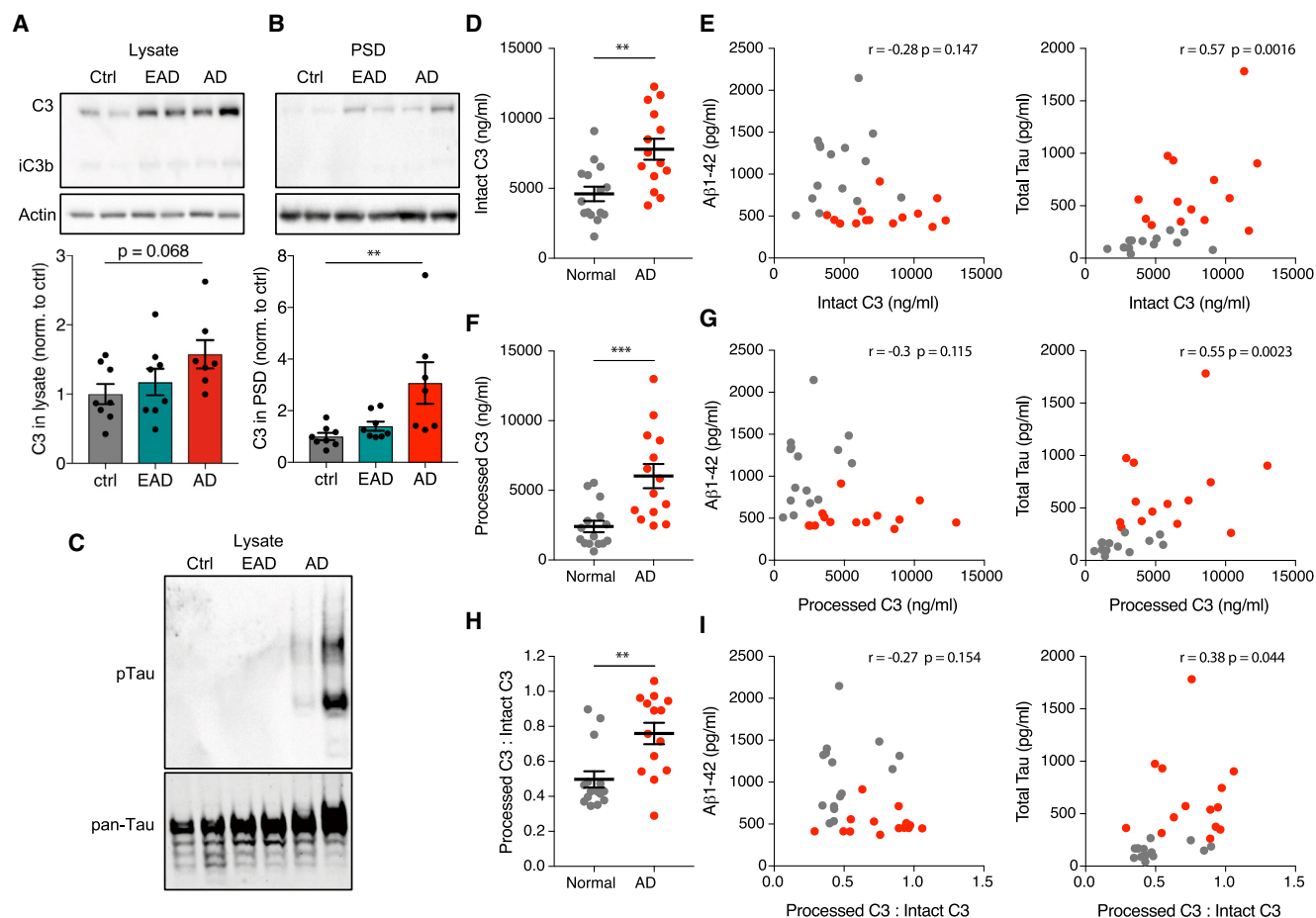


Figure 7. Elevated Synaptic C3 Levels in AD Patient Brains and Elevated Levels and Processing of C3 in AD Patient CSF

(A and B) Representative immunoblots and quantification of relative C3 levels in total lysates (A) and isolated PSD fractions (B) from control, early AD, and AD patient brains. C3 levels were normalized to the average control level. $n =$ tissue from 7–8 individuals per group. $**p < 0.01$. One-way ANOVA with Tukey's post hoc test. All data are presented as mean $\pm$ SEM.

(C) Representative immunoblot of phospho-tau (AT8) and total tau in lysates from control, early AD, and AD patient brains.

(D) Intact C3 concentration in CSF from cognitively normal and AD patients.

(E) Correlation of C3 and Aβ1-42 or tau concentration in CSF from cognitively normal and AD patients.

(F) Processed C3 concentration in CSF from cognitively normal and AD patients.

(G) Correlation of processed C3 and Aβ1-42 or tau concentration in CSF from cognitively normal and AD patients.

(H) Ratio of processed/intact C3 in CSF from cognitively normal and AD patients.

(I) Correlation of processed/intact C3 ratio and Aβ1-42 or tau concentration in CSF from cognitively normal and AD patients. Each dot represents data from an individual (gray, cognitively normal; red, AD).

$**p < 0.01$, $***p < 0.001$. Unpaired t test. The Pearson correlation coefficient and t test were used for correlations. See also Figure S7 and Table S1.

One finding is that C3KO partially rescued neurodegeneration and electrophysiological and behavioral readouts in TauP301S mice without significantly altering gliosis and tau pathology. This suggests complement is downstream of tauopathy and activation of microglia and astrocytes in driving neurodegeneration and brain dysfunction, and it shows that blocking complement can provide a benefit even in the face of ongoing glial activation. The incomplete rescue likely reflects that in addition to triggering deleterious glial activation, pathological tau mediates complement-independent neurotoxicity, probably by affecting multiple molecular pathways within neurons (Dejanovic et al., 2018; Guo et al., 2017; Ittner and Ittner, 2018). Deletion of

the microglial C3aR1 receptor in TauP301S mice was shown to abolish astrocyte C3 expression, curtail tau pathology, and normalize hippocampal-dependent memory (Litvinchuk et al., 2018). In contrast to C3KO, C3aR1-KO dramatically diminished activation of microglia and astrocytes in TauP301S mice, suggesting potential C3-independent effects of C3aR1-KO. Previous work has identified activated astrocytes as neurotoxic and has identified these astrocytes in neurodegenerative disease patient brains using C3 staining as a marker of activation (Liddelow et al., 2017). While release of an unknown toxic factor or factors by A1 astrocytes has been proposed (Liddelow et al., 2017), our results suggest that the expression of classical complement

pathway components by activated astrocytes (and microglia) could be sufficient to mediate neurotoxicity. Consistent with the notion that tauopathy plays a key role in driving neurodegeneration via glial activation (including induction of complement), progression of degeneration and cognitive symptoms in AD are correlated with tauopathy, rather than amyloidosis. Furthermore, whereas amyloid plateaus early in disease, gliosis increases throughout disease and positively correlates with phospho-tau (Serrano-Pozo et al., 2011); in addition, positron-emission tomography (PET) imaging shows glial activation correlates more strongly with tau aggregation than amyloid depositions in AD patients (Dani et al., 2018).

A key finding of our study is the increased levels and activation of C3 in PSD fractions and CSF from AD patients. The enriched C3 protein in PSD fractions from AD patients, particularly at a stage of disease with tau pathology, is consistent with C3 being upregulated and deposited at synapses in response to tau pathology. We previously reported similar results for the upstream complement component C1q (Dejanovic et al., 2018). The detection of increases of both the overall C3 levels and the fraction of C3 that has been processed in AD patient CSF provides additional evidence for complement induction and activation in AD brains. C3 levels are elevated, whereas C1q protein levels in CSF from AD patients are reduced (Khoonsari et al., 2016; Lui et al., 2016), potentially reflecting C1q deposition in the brain, which could contribute to the enhanced processing of C3 detected in CSF. The correlation of intact and processed C3 levels in CSF with tau suggests that elevated complement expression and activation in AD brains may especially be driven by tau pathology.

Our results implicating the classical complement pathway in not just response to amyloidosis but also response to tauopathy fit with a growing body of evidence linking complement activation to AD and other neurodegenerative diseases. Using an unbiased proteomic approach, we have identified C1q as one of the most highly enriched proteins in synapses in tauopathy mice (Dejanovic et al., 2018). Besides AD, activation of the classical complement pathway could be a cause of synapse loss and neurodegeneration in other diseases, including progranulin-deficient frontotemporal dementia (Lui et al., 2016), virus-induced memory impairment (Vasek et al., 2016), glaucoma (Stevens et al., 2007), CNS lupus (Bialas et al., 2017), schizophrenia (Sekar et al., 2016), and neuroinflammation after stroke (Alawieh et al., 2018). Consistent with deleterious roles for complement in other CNS diseases, our demonstration of neuroprotection by genetically eliminating complement in a tauopathy and AD mouse model and identification of increased C3 abundance and pathway activation in AD patient samples points to a harmful role of complement in AD. Given the correlations of complement induction and neurodegeneration with tau pathology in mouse models and patients, reducing complement could be a potential therapeutic strategy for treating AD, especially at stages of disease driven by tau pathology.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.celrep.2019.07.060>.

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AUTHOR CONTRIBUTIONS

T.W., B.D., V.D.G., A.G., R.E., K.S., Y.W., T.-M.W., M.H., and J.E.H. performed experiments. S.S., M.A.H., K.H.B., M. Stark, H.N., and O.F. analyzed data. T.W., B.D., S.S., W.J.M., J.E., M.C.C., D.V.H., R.A.D.C., M. Sheng, and J.E.H. designed and interpreted experiments. T.W., B.D., and J.E.H. wrote the manuscript. All authors read and edited the manuscript.

DECLARATION OF INTERESTS

All authors are current or former employees of Genentech.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
C1q (4.8)	Abcam ab182451	RRID: AB_2732849
C3 (11H9)	Novus Biologicals NB200-540	RRID: AB_10003444
GFAP (ASTRO6)	Thermo Scientific MS-1376-P	RRID: AB_62808
Iba1	Wako Chemicals 019-19741	RRID: AB_839504
LAMP1 (1D4B)	Abcam ab25245	RRID: AB_449893
Synapsin	Synaptic Systems 106006	RRID: AB_2622240
CD68 (FA-11)	Bio-Rad MCA1957	RRID: AB_322219
Campbell-Switzer Alzheimer Silver stain (NSA)	NeuroScience Associates	https://www.neuroscienceassociates.com/technologies/alzheimers-pathology-stain/
Amino-Cupric Silver stain (NSA)	NeuroScience Associates	https://www.neuroscienceassociates.com/technologies/disintegrative-degeneration-stain/
NeuN (NSA)	Millipore MAB337B	RRID: AB_177621
AT8 (NSA)	ThermoScientific MN1020B	RRID: AB_223648
GFAP (NSA)	Dako Z0334	RRID: AB_10013382
Iba1 (NSA)	Wako 019-19741	RRID: AB_839504
anti-C3a	R&D Systems AF3677	RRID: AB_2066614
biotinylated anti-C3	Cappel Research Reagents	55033
anti-iC3b/C3b (7C12)	Kerafast	EG1004
biotinylated anti-C3 F(ab)	Protos Immunoresearch	265
pTau (AT8)	ThermoFisher MN1020	RRID: AB_223647
pan-Tau	Synaptic Systems 314003	RRID: AB_993039
C3	Abcam [EPR19394]	ab200999
Beta-Actin HRP-conjugate (AC-15)	Sigma A3854	RRID: AB_262011
Biological Samples		
Human CSF cohort 1	Folio Biosciences	n/a
Human CSF cohort 2	PrecisionMed	n/a
Frozen human brain tissue	Folio Biosciences	n/a
Chemicals, Peptides, and Recombinant Proteins		
Paraformaldehyde	Electron Microscopy Sciences	15710
ProLong Diamond Antifade Mountant with DAPI	Invitrogen	P36971
Normal Donkey Serum	Abcam	ab7475
NSA Mounting Solution pH 6.0	NeuroScience Associates	https://www.neuroscienceassociates.com/instructions/mounting/
Methoxy-X04	Tocris Bioscience	4920
Critical Commercial Assays		
Simoa	Quanterix	https://www.quanterix.com/products-technology/simoa-assay-kits
Bio-Plex Pro Mouse Cytokine 23-Plex Assay	Bio-Rad	#m60009rdpd
Deposited Data		
RNaseq PS2APP sorted cells, 7, 13 mo	(Friedman et al., 2018)	GSE75431
RNaseq TauP301S sorted microglia	(Friedman et al., 2018)	GSE93180
RNaseq PS2APP sorted cells, 11.5 mo	This study	GSE129770
RNaseq TauP301S sorted neurons and astrocytes	This study	GSE129797

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Models: Organisms/Strains		
MOUSE: Thy-1.PrP.hu.APP.hu.PS2.tg.B6	Roche	(Richards et al., 2003)
MOUSE: Prnp.Tau.P301S.tg.B6N	The Jackson Laboratory	Stock No: 02481
MOUSE: C3.ko.B6N	Washington University, St. Louis, MO	(Circolo et al., 1999)
Software and Algorithms		
MATLAB	Mathworks	https://www.mathworks.com/
Prism 7	GraphPad	https://www.graphpad.com/scientific-software/prism/
ImageJ	NIH	https://imagej.nih.gov/ij/
Imaris 8.3.1	Bitplane	https://imaris.oxinst.com/
Image Lab Software	Bio-Rad	http://www.bio-rad.com/en-us/product/image-lab-software?ID=KRE6P5E8Z
Image Studio Software	LI-COR	https://www.licor.com/bio/image-studio/
Photobeam Activity System - Open Field	San Diego Instruments	https://sandiegoinstruments.com/product/pas-open-field/
TopScan	CleverSys	http://cleversysinc.com/CleverSysInc/csi_products/topscan-suite/

LEAD CONTACT AND MATERIALS AVAILABILITY

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Jesse Hanson (hanson.jesse@gene.com). This study did not generate new unique reagents

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

PS2APP mice express human APP with the K670N/M671L mutation and human presenilin 2 with the N141I mutation, driven by the Thy1 and PrP promoter, respectively (Richards et al., 2003), and all experiments in this study used homozygous PS2APP mice and non-transgenic littermates as controls. TauP301S mice express human Tau with the P301S mutation, driven by the PrP promoter (Yoshiyama et al., 2007), and all experiments in this study used heterozygous TauP301S mice and non-transgenic littermates as controls. For dendritic spine analysis, PS2APP mice were crossed to the Thy1-GFP-M line transgenic mice, which express EGFP in a subset of neurons (Feng et al., 2000). These disease models were crossed to C3KO mice, which were generated by replacing the 5'-flanking region of the C3 gene with the neomycin-resistance gene resulting in lack of functional protein expression (Circolo et al., 1999). Adult male and female mice were used in experiments with information on ages and sex used in each specific experiment provided in the figure legends. Experimenters were blind to genotype for all behavioral measurements, electrophysiological, microscopic, and histological analyses. All animal studies were authorized and approved by the Genentech Institutional Animal Care and Use Committee.

Human subjects

Post mortem brain tissue (superior frontal gyrus) was obtained from patients with putative AD as indicated by cognitive evaluation scores ("EAD") or pathology-confirmed AD ("AD") along with age-matched controls from Folio Biosciences and Precision Medicine. Human samples were procured with Ethics Committee approval and written informed consent. CSF from AD patients was also obtained from Folio Biosciences and was collected as previously described with exclusion of samples with evidence of blood contamination (Wildsmith et al., 2014). Adult male and female patients were used in this study and demographic data can be found in Table S1.

METHOD DETAILS

FACS of brain cells for RNA sequencing

TauP301S microglia sorting was previously described (Friedman et al., 2018) and neurons and astrocytes were sorted from those same mice using published protocols (Srinivasan et al., 2016). In brief, mice were perfused and their hippocampi dissected and dissociated at 4°C. Cells were fixed, immunolabeled, and FACS sorted, using NeuN, GFAP, and CD11b, to isolate neurons, astrocytes, and

microglia, respectively. cDNA was generated using Nugen's RNA-Seq method for low-input RNA samples, Ovation RNA-Seq System v2 (NuGEN), and cDNA was further processed using Illumina's TruSeq RNA Sample Preparation Kit v2 (Illumina). The previously published datasets are available as GSE75431 (PS2APP sorted astrocytes, neurons and microglia, 7 and 13 months) and GSE93180 (TauP301S sorted microglia). Previously unpublished data for sorted astrocytes, neurons, and microglia from 11.5-month-old PS2APP mice is available as GSE129770, and previously unpublished data for astrocytes and neurons from TauP301S mice is available as GSE129797.

Read processing and mapping

The fastq sequence files for all RNA-seq samples were filtered for read quality (keeping reads where at least 70% of the cycles had Phred scores ≥ 23), and ribosomal RNA contamination. The remaining reads were aligned to the mouse reference genome (GRCm38) using the GSNAP alignment tool (Wu and Nacu, 2010). Alignments were produced using the following GSNAP parameters: "-M 2 -n 10 -B 2 -i 1 -N 1 -w 200000 -E 1 -pairmax-rna=200000 -clip-overlap." These steps, and the downstream processing of the resulting alignments to obtain read counts and RPKMs per gene (over exons of RefSeq gene models) are implemented in the Bioconductor package, HTSeqGenie (v 4.10.0). Only uniquely mapped reads were used for further analysis.

Gene set score analysis

Gene set scores were calculated as previously described (Friedman et al., 2018). Briefly, for a set of genes, RPKMs were log2 transformed, and then centered by subtracting the mean of the log2 RPKMs for the control (WT or vehicle) samples for each animal. This represents the log2 fold change for each sample, with respect to the mean of the control samples. The gene set score per sample was then calculated by taking the average log2 fold change over all the genes in the gene set, for a given sample. The astrocyte activation gene sets were obtained from Tables 1 and 2 in Zamanian et al. (2012), with minor manual curation. Briefly, starting with the two lists of the top 50 genes whose expression changed in MCAO, or LPS, we took the intersection of these two gene lists to form the pan-reactive gene set. The remaining LPS genes that were not included in the pan gene set are denoted A1, and the remaining MCAO genes not included in the pan gene set, are denoted A2. In addition, C3 was included in the A1 gene set and Stat3 in the A2 gene set based on their inclusion in gene sets in Clarke et al. (2018).

Tissue processing and immunohistochemistry

Mice were deeply anesthetized and transcardially perfused with phosphate-buffered saline (PBS). Hemi-brains were drop fixed for 48h at 4°C in 4% paraformaldehyde. After being cryoprotected and frozen, up to 40 hemi-brains were embedded per block in a solid matrix and sectioned coronally at 30 μ m (MultiBrain processing by Neuroscience Associates, NSA) before being mounted onto slides. TauP301S brain sections were stained for AminoCu, AT8, GFAP, Iba1, and NeuN, and PS2APP brain sections were stained for AminoCu, Campbell-Switzer, GFAP, and Iba1 at NSA using established protocols. Additional immunostainings were performed on free-floating sections in PBS with 0.1% Triton X-100 (PBST), blocked with 5% normal donkey serum in PBST (NDST), and incubated overnight at 4°C with primary antibodies in 1% NDST. Secondary antibodies in 1% NDST were incubated for 2-3h at room temperature, washed in PBST and PBS and mounted using NSA Mounting Solution pH 6.0 (NeuroScience Associates). Slides were coverslipped with ProLong Diamond Anti-fade Mountant with DAPI. Primary antibodies: C1q (1:1000 clone 4.8, rabbit monoclonal, Abcam ab182451), C3 (1:500 clone 11H9, rat monoclonal, Novus Biologicals NB200-540), GFAP (1:1000 clone ASTRO6, mouse monoclonal, Thermo Scientific MS-1376-P), LAMP1 (1:500 clone 1D4B, rat monoclonal, Abcam ab25245), CD68 (1:1000, clone FA-11 rat monoclonal, Bio-Rad MCA1957), Synapsin (1:500, chicken polyclonal, Synaptic Systems 106006), and Iba1 (1:1000, rabbit polyclonal, Wako Chemicals 019-19741). Alexa Fluor 594 and 647 secondary antibodies were used at 1:500. To probe for amyloid β , slides were incubated in 10 μ M Methoxy-X04 (Tocris Bioscience 4920) in 40% ethanol/60% distilled water and then differentiated in 0.2% NaOH in 80% EtOH before coverslipping.

Whole slide imaging and analysis

Immunofluorescent slides were imaged at 200x magnification using the Nanozoomer-XR (Hamamatsu Corp, San Jose, CA) whole slide scanner equipped with a fluorescent imaging module and standard filter wheel. All whole slide image analysis was performed in a blinded manner using MATLAB v9.4 (Mathworks, Natick, MA). Total tissue area was detected by thresholding on the DAPI and Alexa-594 signal and merging and processing of the binary masks by morphological operations. Hippocampal regions of interest were marked up manually. Pixel intensity was evaluated in 8-bit grayscale and the C1q or C3 (Alexa-594) integrated pixel intensity in the whole brain section or hippocampus was normalized to the whole tissue or hippocampal area, respectively. GFAP (Cy5), Iba1 (Alexa-594), LAMP1 (Alexa-594) and methoxy-X04 (DAPI) plaque staining was analyzed using a top-hat filter and local threshold followed by morphological opening and closing. Astrocytic C3 integrated intensity was calculated from the area defined by the GFAP mask. For LAMP1 and methoxy-X04 staining, shape factor, roundness and solidity features were used to eliminate elongated objects. In addition, a minimum size of 34 μ m² was applied to exclude small areas of staining. The detected plaques were used as markers in a marker-controlled watershed segmentation to create watershed lines of separation. The plaque mask was then dilated by 13.8 μ m but constrained to be within watershed lines to prevent merging of plaques in close proximity during dilation. LAMP1 positive staining was normalized to the whole tissue area. LAMP1, GFAP, and Iba1 plaque-associated staining (constrained to be within the dilated plaque mask) was normalized to dilated plaque area. For the PS2APP model, astrocytic C3 plaque-associated integrated

intensity (constrained to be within the dilated plaque mask) was normalized to the dilated plaque area. Data was averaged from two sections per animal. In a small number of cases, out of focus sections and sections with region of high tissue background were manually excluded. Brightfield slides processed by NeuroScience Associates were imaged on the Leica SCN400 whole slide acquisition system (Leica Microsystems, Buffalo Grove, IL) at 200x magnification. Quantification of chromogenic staining (AminoCu, NeuN, GFAP, Iba1, AT8, Campbell-Switzer plaque) area was performed using grayscale and color thresholds (Brey et al., 2003) followed by morphological operations. Positive stain area was normalized to the whole brain section or manually marked up hippocampal area.

Analysis of dendritic spine density relative to amyloid plaques

The somatosensory cortex of PS2APP mice was imaged *ex vivo* via 2-photon microscopy. Twenty-four hours before brain harvesting, animals were injected with 10 mg/kg Methoxy-X04 intraperitoneally to visualize individual plaques (Klunk et al., 2002). Animals were euthanized and transcardially perfused with PBS, and brains were harvested, incubated for 48 hours in 4% PFA, and then washed and embedded in agarose for imaging. Apical dendrites and their spines in somatosensory cortex were imaged via a two-photon laser-scanning microscope (Ultima *In Vivo* Multiphoton Microscopy System; Prairie Technologies) using a Ti:sapphire laser (MaiTai DeepSee Spectra Physics; Newport) tuned to 840 nm and a 60x numerical aperture 1.0 objective lens (Olympus) with pixel resolution of 0.1 μm /pixel across a 1024×1024 pixel field of view (FOV) using 0.5 μm steps. At least five cells were collected per condition per animal. For plaque density measurements, larger image stacks were collected using a 20x objective across a 1024×1024 pixel FOV with 2 μm steps. Dendritic spine density and size measurements were generated using custom, semiautomated image analysis routines in MATLAB (MathWorks). Spine density was estimated as the total number of visible dendritic spines divided by the corresponding length of dendrite. Relative spine volumes were estimated for each detected spine based on the number of corresponding GFP+ pixels in x, y, z dimensions above a local threshold applied as part of an automated image segmentation algorithm. For comparison of spine density relative to plaques in PS2APP animals, an FOV containing a dendrite and nearby plaque within 20 μm was considered “near plaque” and an FOV containing only a dendrite with no visible plaque was considered “away from plaque.” To meet the “away from plaque” criteria, we confirmed that no plaque was present in the FOV and at least 100 μm outside of the containing FOV. Plaque density was assessed over the same region as spine density measurements. Larger volumes (~ 200 μm depth) were collected and plaque density was quantified by a threshold-based MATLAB routine designed to automatically identify methoxy-X04-labeled plaques. Imaging and analysis were performed under blinded conditions.

Brain lysate cytokine measurements

Following perfusion with PBS. Hippocampi were isolated and homogenized in cold RIPA buffer (50 mM Tris-HCl, 150 mM NaCl, 2 mM EDTA, 1% NP-40, 0.1% SDS) supplemented with phosphatase and protease inhibitors, using a TissueLyser (2 $\times$ 30 Hz, 3 min at 4°C; QIAGEN). After homogenization, samples were centrifuged at 20,000 g for 20 min and the supernatants were transferred into new tubes. The protein concentration in each sample was measured by BCA (ThermoFisher). Cytokine levels in the samples were measured using a Bio-Plex Pro Mouse Cytokine 23-Plex Assay (Bio-Rad) and normalized to total protein concentration.

Longitudinal vMRI imaging

MRI was performed on a 7T Bruker (Billerica, MA) system with a 4-channel receive-only cryogen-cooled surface coil and a volume transmit coil (Bruker, Billerica, MA). T2-weighted images were acquired with a multi-spin echo sequence: TR 5100 ms, TE 10, 20, 30, 40, 50, 60, 70, 80ms, 56 contiguous axial slices of 0.3 mm thickness, FOV 19.2 mm $\times$ 19.2 mm, matrix size 256 $\times$ 128, 1 average, with a scan time of 11 min/mouse. During imaging, anesthesia for mouse was maintained at 1.5% isoflurane and rectal temperature was maintained at $37 \pm 1^\circ\text{C}$ using a feedback system with warm air (SA Instruments, Stony Brook, NY). Equal number of males and females were included to detect any gender difference. The regional and voxel differences in the brain structure were evaluated by registration-based region of interest (ROI) analysis. In brief, multiple echo images were averaged and corrected for field inhomogeneity to maximize the contrast to noise ratio and the images were analyzed based on a 20-region pre-defined *in-vivo* mouse atlas (<http://brainatlas.mbi.ufl.edu/>) that was co-registered to a study template and warped to individual mouse datasets. All the co-registration steps were performed in SPM8 (Wellcome Trust Centre for Neuroimaging, UCL, UK). The ROI volumes were normalized to whole brain volumes to minimize any genotype differences associated with size of the animals.

Synapse engulfment assay

Microglia engulfment analysis was performed as previously described (Dejanovic et al., 2018). Briefly, immunostained brain sections were imaged on a Zeiss LSM 710 confocal microscope taking up to 35 z stacks (0.4 μm steps size) with a 100x oil objective. CD68-positive structures from single microglia were 3D-reconstructed using the surface rendering function in Imaris 9.2 and synapsin puncta inside the lysosomes were quantified using the spots function. Synapsin puncta within 7-8 microglia in the CA1 region were analyzed per mouse.

Electrophysiology

Population spike LTP was recorded as previously described (Hanson et al., 2013). Briefly, horizontal hippocampal slices (400 μm) were prepared with a vibrating sectioning system (Leica, Germany), and were recorded in oxygenated artificial cerebrospinal fluid

(ACSF) containing (in mM) 127 NaCl, 2.5 KCl, 1.3 MgSO₄, 2.5 CaCl₂, 1.25 Na₂HPO₄, 25 NaHCO₃, and 25 glucose. Slices were prepared in ice-cold oxygenated ACSF with the MgSO₄ concentration elevated to 7 mM, NaCl replaced with 110 mM choline, and with 11.6 mM Na-ascorbate and 3.1 mM Na-pyruvate. EPSPs and PSs were measured from the stratum radiatum and stratum pyramidale of CA1, respectively, in response to electrical stimulation of Schaffer collateral inputs. LTP was induced using 1, 2, or 3 bursts of stimulation consisting of 5 pulses at 100 Hz, separated by 20 s, each.

Open field and Morris water maze

Spontaneous locomotor activity was measured with an automated Photobeam Activity System-Open Field (San Diego Instruments). Mice were placed individually in a clear plastic chamber (41L × 41W × 38H cm), and their horizontal and vertical movements were monitored for 15 min/session with two 16 × 16 photo-beam arrays.

Mice were trained in the Morris water maze as previously described (Hanson et al., 2014). Briefly, mice were trained to locate a hidden platform (15 cm diameter) submerged 1.5 cm below the water for five consecutive days. Mice received two training sessions per day, with each session consisting of three trials (10 min between trials), and sessions separated by 3–4 hours. The platform location remained constant throughout hidden-platform training, but the drop location was changed between trials. Mice that did not find the platform within 60 s were guided to it and placed on it for 10 s. A spatial probe trial was performed at the beginning of day 4 (probe 1) and at 24h and 72 hr (probe 2 and 3) after the completion of training. During the probe trial, the platform was removed and mice were allowed to swim in the pool for 60 s. Escape latencies, swim paths, swim speeds, percentage of time spent in each quadrant, and platform crossings in a zone 200% the size of the platform were recorded using a ceiling-mounted camera and tracked using CleverSys TopScan software.

Human brain lysates and biochemistry

PSD fractions were isolated as described (Dejanovic et al., 2018; Peng et al., 2004). Briefly, pieces of brain tissue were homogenized in homogenization buffer (5 mM HEPES (pH 7.4), 1mM MgCl₂, 0.5mM CaCl₂) supplemented with phosphatase and protease inhibitors using a Teflon homogenizer. After 1,400 g, 10 min centrifugation, the supernatant was transferred to a new tube and the pellet was re-homogenized and pelleted at 1,400 g, 10 min. Supernatants were combined and centrifugated at 13,800 g, 10 min. Pellet was re-suspended in 0.32 M Tris-buffered sucrose and ultra-centrifuged into 1.2, 1.0, 0.85M sucrose gradient at 82,500 g, 2h. Synaptic fraction was collected between the 1.0 and 1.2M sucrose interface, solubilized with 1% Triton X-100 and centrifuged at 32,800 g for 20 min. The pellet was extracted in 0.5% Triton X-100 and centrifuged at 200,000 g for 1h to yield the final PSD pellet.

CSF total and processed complement assays

C3 and processed C3 were measured in human CSF using custom single molecule array (Simoa) assays (Quanterix Corp, Boston, MA, USA). For the C3 assay, the main reagents consisted of paramagnetic carboxylated beads (Quanterix Corp, Boston, MA, USA) coated with a goat anti-C3a antibody (AF3677, R & D Systems, Minneapolis, MN, USA) and a biotinylated goat anti-C3 detection antibody (55033, Cappel/MP Bio). For processed C3, which measures the iC3b/C3b and C3c protein fragments, the main reagents consisted of paramagnetic carboxylated beads (Quanterix Corp, Boston, MA, USA) coated with a mouse anti-iC3b/C3b antibody (EG1004, distributed by Kerafast, Boston, MA, USA, antibody from the lab of Ronald P. Taylor, PhD, University of Virginia) and a biotinylated goat anti-human C3 F(ab) detection (265, Protos Immunoresearch, Burlingame, CA, USA). Conjugations were done using the standard recommended concentrations and challenge ratios from Quanterix. Please note that the raw detection antibody for C3 was protein G purified before use in conjugation, as the raw antibody consists of an antiserum.

Assays were run using one of the standard protocols for the Simoa HD-1 instrument from Quanterix. In the protocol 25 μ L of capture coated beads were incubated for 30 minutes with 25 μ L of diluted sample. After washing, immunocomplexes were incubated for 5 minutes with 100 μ L of the biotinylated detection antibody. Washed immunocomplexes were incubated for 5 minutes with 100 μ L of streptavidin-conjugated β -galactosidase (Quanterix). After a last round of washes, the beads were re-suspended in resorufin β -D-galactopyranoside (Quanterix) and the mixture was then applied to Simoa discs. The HD-1 analyzer was used to read the resulting fluorescent signal and calculate the average number of enzymes per bead (AEB) for tested samples. The reported AEB values were analyzed against a calibrator curve constructed by AEB measurements on native human C3 or C3b protein (Complement Technology) serially diluted in assay diluent. Samples were analyzed using a single batch of reagents and testing across three runs. For each of the three runs, the PC3 and C3 assays were run together with calibrators and controls for each assay and 5 normal and 5 Alzheimer's patient samples tested at the chosen dilutions.

Western blotting

Immunoblotting was performed as recently described (Dejanovic et al., 2018). Briefly, protein samples were boiled in reducing SDS loading buffer and ran on NuPAGE gradient gels (Invitrogen). After transfer to nitrocellulose membranes (Bio-Rad), membranes were blocked with milk or BSA and incubated with primary antibodies overnight, followed by secondary HRP-conjugated (abcam) or fluorescent (Li-Cor) antibodies. Immunoreactivity was detected on ChemiDoc XRS+ (Bio-Rad) or Odyssey infrared Imaging System (Li-Cor).

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantification is described in Method Details and in figure legends. Number of samples is shown in the figure and described in the figure legends. Statistical analysis was performed using GraphPad Prism 7 and the used statistical tests are defined in the figure legends. Data throughout the paper is displayed as mean $\pm$ SEM, and statistical significance is defined as: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. No outlier analysis was performed or data points excluded. Sample size was chosen according to that used for similar experiments in previously published literature. Experimenters were blinded whenever possible to experimental condition during data acquisition or quantification.

DATA AND CODE AVAILABILITY

The RNaseq data analyzed in this study have been deposited to NCBI's Gene Expression Omnibus as Accession Numbers GSE75431, GSE93180, GSE129770, and GSE129797.